Electromagnetic Wave Modes in a Rectangular Waveguide

In this example we want to investigate the propagation modes of electromagnetic waves inside a rectangular waveguide. Consider a perfect conductor rectangular box extended in z direction (Fig.1). We want to find different wave modes which propagate in the z direction, so variation of fields in the z direction is in the harmonic form $e^{ikz-iwt}$. In this configuration geometrical constraints of the problem appear in the form of boundary conditions. Boundary condition for a perfect conductor requires that there is no tangential electric field on the boundaries.

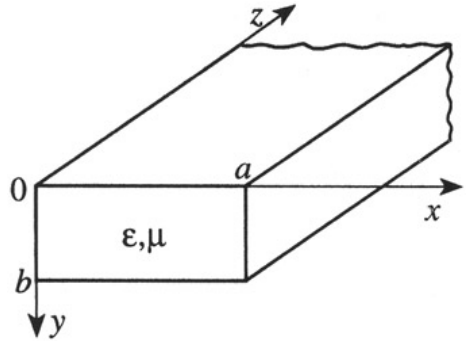

Figure 1: Schematic of a rectangular waveguide and its coordinates

As the first observation let's examine if it is possible to have a pure transverse wave propagating in the waveguide or not. For instance take electric field to be transverse and in y direction. To satisfy boundary conditions it needs to be zero at $x = 0$ and $x = a$. This means that y component of $\mathbf{E}$ (Electric Field) must have variation in x direction and so its curl has some non-zero component in z direction. Now considering the harmonic form of Faraday equation, $\nabla \times \mathbf{E} = iw\mathbf{B}$, we can see that in this case $\mathbf{B}$ needs to have a component in the direction of propagation z. Generally, we can conclude that there is no pure transverse propagating mode in the waveguide. Therefore, we can categorize all of modes of propagation in those with transverse electric field and those with transverse magnetic field. Technically the first ones are called TE modes and the latter ones TM modes. Of course in practice we can have any linear combination of theses modes propagating in a waveguide.

1 TE Modes

Following the same way as did in discussion class, we just consider the simplest TE mode, for which $\mathbf{E}$ is in y direction and varies only in x direction. From boundary conditions we know that $\mathbf{E}$ must be zero at $x = 0$ and $x = a$. Therefore,

$$E_y = E_0(x)e^{ikz-iwt},$$

where $E_0(0) = E_0(a) = 0$, and from famous Fourier expansion we know that E_0 must be a linear combination of $\sin \frac{m\pi x}{a}$ for integer m. Each of these harmonics corresponds to a TE mode. The lowest one is for $m = 1$. Now we are going to find the dispersion relation for this lowest mode. First let's write down harmonic form of Maxwell's equations in the absence of sources:

$$\nabla \times \mathbf{E} = iw\mathbf{B} \tag{1}$$

$$\nabla \times \mathbf{B} = -\frac{iw}{c^2}\mathbf{E} \tag{2}$$

Taking curl from the Eq.1 we get

$$\nabla(\nabla \cdot \mathbf{E}) - \nabla^2\mathbf{E} = \nabla \times (iw\mathbf{B}). \tag{3}$$

Now using Eq.2 at the right hand side and taking into account the absence of sources which means $\nabla \cdot \mathbf{E} = 0$, we will have

$$-\nabla^2\mathbf{E} = \frac{w^2}{c^2}\mathbf{E}. \tag{4}$$

Plugging in the form of first TE mode into Eq.4 and noting that

$$\nabla^2\mathbf{E} = \frac{\partial^2\mathbf{E}}{\partial x^2} + \frac{\partial^2\mathbf{E}}{\partial z^2} = -\left[\left(\frac{\pi}{a}\right)^2 + k^2\right]\mathbf{E},$$

we get

$$\frac{w^2}{c^2} = k^2 + \left(\frac{\pi}{a}\right)^2, \tag{5}$$

which is called *dispersion relation* of the mode. It specifies what frequency we have for each wavevector (or equivalently wavelength). Immediately from this dispersion relation (Eq.5) we can see that there is a minimum value $w = c\frac{\pi}{a}$ for w, which means it's impossible to have a propagating wave below this frequency in this mode, so this frequency is usually called *cut-off frequency.* As it's clear both from the dispersion expression (Eq.5) and its plot (Fig. 2) this minimum happens at $k = 0$. By increasing k when $k \gg \frac{\pi}{a}$ we can see that the relation between k and w becomes similar to the famous simple dispersion relation of plane electromagnetic wave in free space, $w = ck$.

We should be aware that k does not represent all spatial variation rate of the fields. It just shows harmonic variation of the fields in the direction of propagation z. But we know there is also a transverse variation length which is proportional to the waveguide size a, that has the form $\sin \frac{m\pi x}{a}$. Recalling Faraday equation (Eq.1) we can see that actually size of longitudinal and transverse components of magnetic field are proportional to these rate of variation, $\frac{B_x}{B_z} \propto \frac{ka}{\pi}$, which suggest that for large ka when the variation length in z direction is very smaller than a the wave looks more directed along the z direction

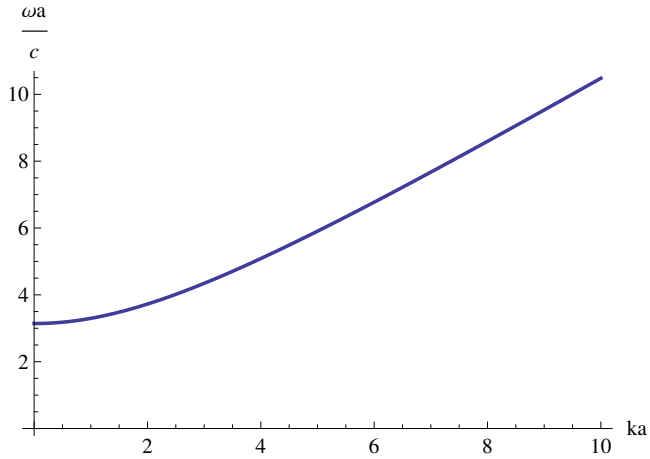

Figure 2: Dispersion relation for the first TE mode

and when it is relatively small, pointing vector of the wave makes some considerable angle with z direction. Qualitatively, we can say that in the latter limit the wave is bouncing off from the side of the waveguide as it propagates and this is the origin of the deviation of dispersion relation in waveguide from its free space form.

2 TM Modes

In the case of TM modes we define our task to find the general form of dispersion relation for these modes. To do this let's start with considering the boundary conditions. Again boundary condition is to set all tangential component of electric field equal to zero on all sides. But since for TM modes electric field always has z component and z component is tangential on both vertical and horizontal sides, we have to choose a suitable form for E_z to vanish at $x = 0$, $x = a$ and also $y = 0$, $y = b$. As we know from Fourier expansion *Sine* harmonics with positive integer number of half wavelength in both x and y directions are such a suitable form:

$$E_{z,mn} = E_{0,mn} \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{ikz-iwt}. \tag{6}$$

In this way we can be sure that E_z is zero at all sides, but we know that electric field can also have x and y components, so before going to find dispersion relation we have to make sure that E_y and E_x are also zero on vertical and horizontal sides respectively. To do that let's find these components for an specific mode with, $\widetilde{E}_{z,mn} = E_{0,mn} \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)$ (where the *tilde sign* on the top of $\widetilde{E}_{z,mn}$ means that it's only the coefficient of $e^{ikz-iwt}$ and we use it just for the sake of brevity). To find E_y use the x component of Eq.1 and y component of Eq.2 and note that $B_z = 0$ and that we can replace $\frac{\partial}{\partial z}$ by ik. Using these facts

we have

$$iwB_x = \frac{\partial E_z}{\partial y} - ikE_y \tag{7}$$

$$ikB_x = -\frac{iw}{c^2}E_y \tag{8}$$

Eliminating B_x from above equations we get

$$ikE_y\left[1 - \frac{w^2}{c^2k^2}\right] = \frac{\partial E_z}{\partial y}, \tag{9}$$

which means $E_y \propto \frac{\partial E_z}{\partial y} \propto \sin\left(\frac{m\pi x}{a}\right)$ and so it satisfies the boundary condition on vertical sides. In the same way, using y component of Eq.1 and x component of Eq.2, we can show that $E_x \propto \frac{\partial E_z}{\partial x} \propto \sin\left(\frac{n\pi y}{b}\right)$ and so it also satisfies its own boundary condition on horizontal sides. Therefore, we can say that choosing E_z as the form introduced in Eq.6 satisfies all boundary conditions.

Now let's find the general form of dispersion relation for TM modes. In the same way that we did for TE mode we just need to use E_z component of Eq.4, noting that for this specific form of E_z (Eq.6) we have

$$\nabla^2 E_{z,mn} = \frac{\partial^2 E_{z,mn}}{\partial x^2} + \frac{\partial^2 E_{z,mn}}{\partial y^2} + \frac{\partial^2 E_{z,mn}}{\partial z^2} = -\left[\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + k^2\right]E_{z,mn},$$

using this in Eq.4 we get the general form of dispersion relation for TM modes:

$$\frac{w^2}{c^2} = k^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2. \tag{10}$$

Here again we can recognize the minimum possible frequency $w_{mn} = c\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$ for each TM mode and it's clear that this cut-off frequency is higher for larger m and n or higher modes. Behaviour of these dispersion relation for $ka \gg m,n$ is again the same as plane wave for the same reason we had for TE modes: variation wavelength in the direction of propagation is very smaller than transverse variation wavelength and so the wave does not feel the boundaries. So these dispersion will have the same plots as the first TE mode (Fig.2) just with different cut-off frequencies.

Extra Comments on The Problem

In the class we had two other explorations on this problem and here I just want to mention them avoiding from going into detail:

Exciting the modes: Recalling that the rate of transition in presence of a dipole moment, $\mathbf{d}$, was proportional to $|\langle f|\mathbf{d}\cdot\mathbf{A}|i\rangle|^2$ and noting that in Coulomb gauge $\mathbf{E} \parallel \mathbf{A}$, we can say that only TM modes can excite dipole transitions for which $\mathbf{d}$ is along the z axis.

Van Hove singularity: Noting that modes in k-space are uniformly distributed, we can see that where $\frac{dw}{dk}$ decreases we have more modes in a tiny small frequency range. Equivalently, we can say that density of state in frequency space is proportional to the inverse of $\frac{dw}{dk}$. Therefore, we can imagine

that we will have singularity when $\frac{dw}{dk} = 0$. This singularity is called Van Hove singularity and clearly we can see such a behaviour in our dispersion relations near $k = 0$.